



11. Worked example: decoupling

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Change of coordinates



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Note on video: The matrix called **D** in the video is not the diagonal matrix of eigenvalues, which is what we call **D** in the text.

Recall that a matrix **A** is **diagonalizable** if it can be written as

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1} \quad \text{where } \mathbf{D} = \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{pmatrix} \tag{4.165}$$

$$\mathbf{S} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}. \tag{4.166}$$

where λ_i and $\mathbf{v}_i$ are corresponding eigenvalue-eigenvector pairs, and the set of eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

Given a diagonalizable matrix **A** the homogeneous system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ can be decoupled as follows. First, define a new vector variable **y** by the equation

$$\mathbf{x} = \mathbf{S}\mathbf{y}, \quad \text{where } \mathbf{S} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}. \tag{4.167}$$

Then substitute this in and rewrite the system in terms of **y**:

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} \\ \mathbf{S}\dot{\mathbf{y}} &= \mathbf{A}\mathbf{S}\mathbf{y} \quad (\text{since } \mathbf{S} \text{ is a constant matrix}) \\ \mathbf{S}\dot{\mathbf{y}} &= \mathbf{S}\mathbf{D}\mathbf{y} \quad (\text{since } \mathbf{A}\mathbf{S} = \mathbf{S}\mathbf{D}) \\ \dot{\mathbf{y}} &= \mathbf{D}\mathbf{y} \quad (\text{multiply by } \mathbf{S}^{-1} \text{ on the left}). \end{aligned}$$

The system in terms of **y** is decoupled, because **D** is a diagonal matrix.

We can then solve for each coordinate function of **y**, and then compute $\mathbf{x} = \mathbf{S}\mathbf{y}$.

and v equal to 1, 0.
What happens to x and y?
You matrix multiply.
[1; 0], 1-- so a1 plus 0, b1 plus 0.
It corresponds to the column vector [a1; b1].
And in the same way, [0; 1] corresponds to [a2; b2] just
by matrix multiplication.
That shows this thing that these correspond.
So in the uv system, the two eigenvectors
are now called i and j.

We can view decoupling as rewriting the system in a new coordinate system in which the basis vectors are the eigenvectors of **A**:

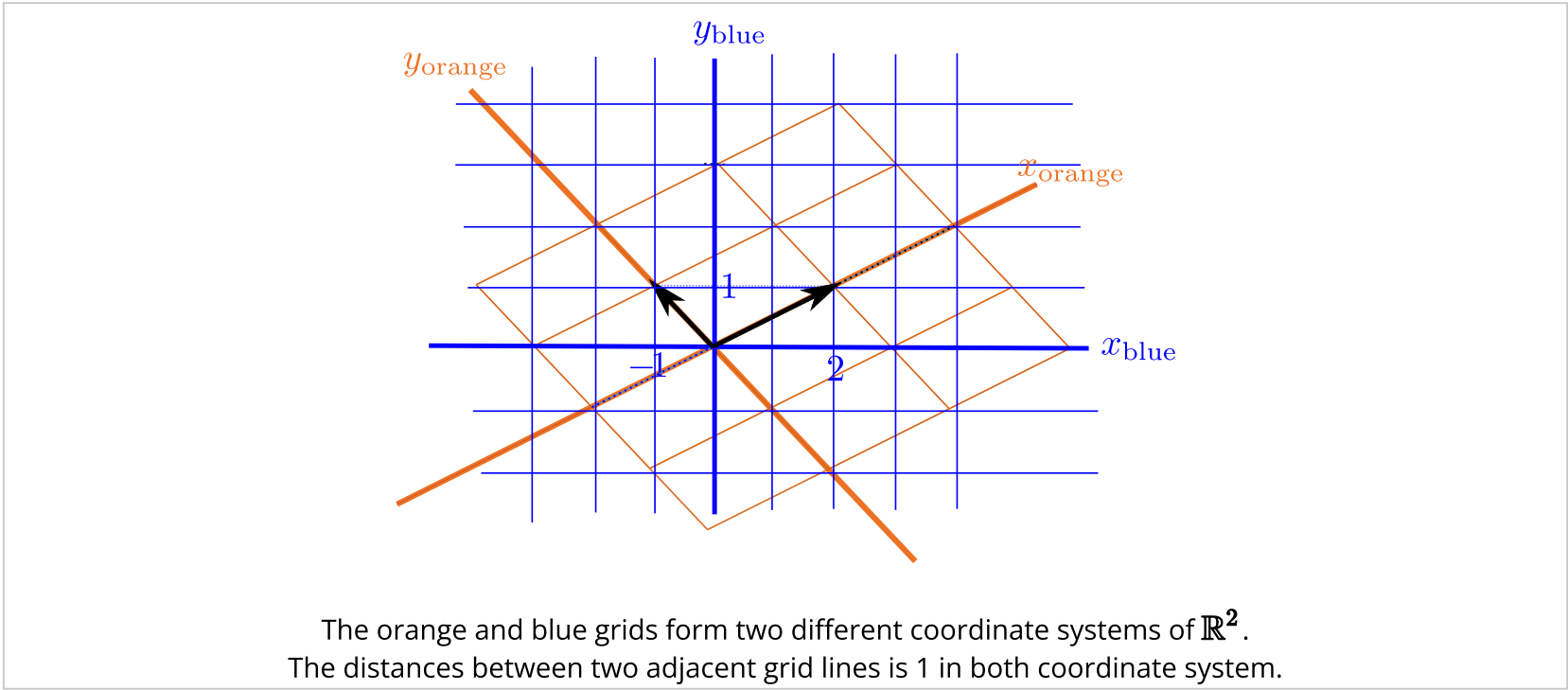
$$\mathbf{x} = \mathbf{S}\mathbf{y}, \quad \text{where } \mathbf{S} = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}.$$

(4.168)

If the eigenvalues are complex, the corresponding eigenvectors will also be complex, i.e., have complex components. All of the above remains formally true, provided we allow all the matrices to have complex entries. This means the new variables **u** and **v** will be expressed in terms of **x** and **y** using complex coefficients, and the decoupled system will have complex coefficients. In most branches of science and engineering, this is perfectly acceptable, and one gets in this way a complex decoupling, and the differential equations can still be solved individually. If one insists on using real variables only, decoupling is not possible with complex eigenvalues. This is the assumption that Professor Mattuck used in the video above.

Change of coordinates

3/3 points (graded)



Consider the vector written as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in the orange coordinate system. This is the vector pointing roughly northeast in the figure above. What is this vector written in the blue coordinate system?

(Enter a single vector as your answer. Enter **[a , c]** for the vector $\begin{pmatrix} a \\ c \end{pmatrix}$. That is, use **comma** to separate the entries, and **square bracket** the entire vector.)

$\begin{pmatrix} x_{\text{blue}} \\ y_{\text{blue}} \end{pmatrix} =$

✓ Answer: [2,1]

Similarly, consider the vector written as $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in the orange coordinate system. This is the vector pointing roughly northwest in the figure above. What is this vector written in the blue coordinate system?

(Enter a single vector as your answer. Enter **[a , c]** for the vector $\begin{pmatrix} a \\ c \end{pmatrix}$. That is, use **comma** to separate the entries, and **square bracket** the entire vector.)

$\begin{pmatrix} x_{\text{blue}} \\ y_{\text{blue}} \end{pmatrix} =$

✓ Answer: [-1, 1]

In general, how do we change from the orange coordinate system to the blue coordinate system? In other words, find the matrix **B** in the following equation:

$$\begin{pmatrix} x_{\text{blue}} \\ y_{\text{blue}} \end{pmatrix} = \mathbf{B} \begin{pmatrix} x_{\text{orange}} \\ y_{\text{orange}} \end{pmatrix}$$

(4.169)

where the **x_orange** and **y_orange** are the coordinates of the vector in the orange coordinate system, and **x_blue**, **y_blue** are the coordinates of the same vector in the blue coordinate system.

(Enter **[[a,b],[c,d]]** for the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. That is, use commas to separate entries within the same row and then enclose the entire row in **square brackets**. Then use **commas** to separate rows, and **square bracket** the entire matrix. Similarly for larger matrices.)

B =

[[2,-1],[1,1]] < Previous

✓ Answer: [[2,-1],[1,1]] Next >

Solution:

We need to read the coordinates in the blue system for the required vectors:

$(\textcolor{brown}{1})$

$(\textcolor{blue}{2})$

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$$\begin{pmatrix} \textcolor{blue}{x}_{\text{blue}} \\ \textcolor{blue}{y}_{\text{blue}} \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \textcolor{brown}{x}_{\text{orange}} \\ \textcolor{brown}{y}_{\text{orange}} \end{pmatrix}$$

(4.170)

Notice the columns of the matrix on the right hand side are the unit basis vectors of the old orange system written in the new blue system.

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